Timing of Preference Submissions under the Boston Mechanism

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Abstract

Exam scores are often the main criteria of students' priorities in college admissions. Depending on the timing of preference submissions relative to the knowledge of scores, students have various degrees of information about their own priorities. I study three widely used timings under the Boston mechanism. The results suggest first, timing that is associated with less information about priorities yields higher expected utility ex-ante; Second, timing with full information about priorities generates full sorting on scores, whereas timing with less information reduces the tension of such sorting.

Keywords: College admissions, school choice, information, Boston mechanism, sorting

JEL Classification: C78, D47, I20, I21, I23

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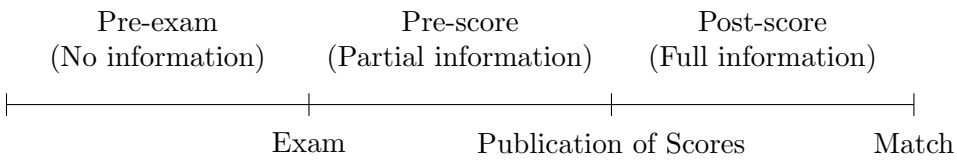
1 Introduction

Assigning students to colleges or schools through centralized procedures has gained popularity in recent decades. Which procedure should be used remains heatedly debated among policy makers and researchers. A large body of literature investigates the choice of specific procedure by primarily considering the complete information environment, that is, students know about the preferences of every other student as well as the priorities when submitting preferences (Abdulkadiroğlu and Sönmez, 2003; Ergin and Sönmez, 2006). Uncertainty is nevertheless widespread.

In this paper, I explore a particular design aspect in a class of exam-based college admission problems: timing to submit preferences. Depending on the choice of timing, students can have different levels of information about their priorities. There are three timings that are frequently used: (1) *post-score* submissions, where students submit preferences after the publication of scores; (2) *pre-exam* submissions, where they have to submit preferences before taking the exam; and (3) *pre-score* submissions, where they submit preferences after taking the exam. The choice of timing varies from one to another in practice. In China, for instance, all three timings are used by different regional clearinghouses (Chen and Kesten, 2017), whereas the pre-score submissions are used in the Irish college admissions (Chen, 2012).

Figure 1: Timing of preference submissions



These observations raise questions about why different timings are used, and what effects they might have. To answer them, I consider a simple setting where students have common ordinal preferences but different cardinal intensities, colleges only value the exam scores of students, and students are matched to colleges using the Boston mechanism. Unlike the strategyproof deferred acceptance (DA) mechanism, the Boston mechanism is known to reward non-truthful behavior by favoring students who top rank a college which is not necessarily their true top choice. The very non-strategyproof element of the Boston mechanism triggers strategic behavior across timings in a tractable way.[1]

[1] In the examples mentioned earlier, not all clearinghouses use the Boston mechanism. In China, the Boston mechanism used to be widespread, which starts to be replaced by a less manipulable mechanism, the parallel mechanism, in recent years (Chen and Kesten, 2017). In Ireland, the Central Application Office (CAO) uses a decentralized version of college-proposing DA mechanism, given that the priority structure may not be unique in this case, truth-telling is not a dominant

Central to my analysis is the application strategies at equilibrium adopted by students under three timings. Post-score and pre-exam submissions are two extreme cases in terms of information availability about scores. Application strategy depends only on scores under post-score submissions, while it hinges entirely on preference intensity under pre-exam submissions. The timing of pre-score submissions is an intermediate case where students obtain some information about their performance. The new feature compared to pre-exam submissions is that for a given preference intensity level, students who receive a better signal are more likely to submit preferences truthfully.

In a main thrust of the paper, I compare welfare for both students and colleges in the game induced by each timing. On the one side, students are weakly better off under pre-exam submissions than under pre-score submissions, and than under post-score submissions from an ex-ante perspective. On the other side, post-score submissions generate full sorting on scores among colleges, and the two other timings create partial sorting since the less good college can compete for some high scoring students who would have been all admitted by the best college under post-score submissions. A closer look at the quality of each college, measured by average score of admitted students, unveils that although both pre-exam and pre-score submissions redistribute high scoring students from the best college to the less good college, only pre-score submissions helps the less good college to improve its quality.

The welfare results shed light on how to organize the timing to submit preferences in centralized college admissions.

First, timing of preference submissions can be considered as an extra design dimension to influence the cardinal welfare for students. Indeed, knowing score under post-score submissions in the Boston mechanism, students ignore their cardinal preferences, in the same vein as in the DA mechanism. In contrast, absent information, students sort credibly according to cardinal intensities, and by doing so, the acceptance chance to the best college rises for students who strongly desire it, while the acceptance chance to the less good college does not fall for students who are relatively more enthusiastic about it.

Second, timing of preference submissions can be employed to tune sorting across colleges. Centralized college admissions usually aim to facilitate application and improve access for students. However, the benefits of such collective effort is often not shared equally among colleges. Assortative matching clearly favors the top colleges, putting the less good colleges in a very difficult situation to compete for good students. Pre-exam and pre-score submissions resolve partially the tension of sorting without harming students from an ex-ante point of view.

strategy for students. However, this game induced by both the parallel mechanism and the college-proposing DA is less tractable.

These findings contribute to the debate on the choice of mechanism by expanding the criteria we should consider. In addition to strategyproofness and stability, policy makers may face other concerns such as ex-ante efficiency as well as sorting. If the desired objective is to reduce the tension of sorting without hurting students (ex-ante), then pre-exam and pre-score submissions under the Boston mechanism are good candidates.

The paper proceeds as follows. Section 2 reviews the literature. Section 3 illustrates the main results of the paper using a numerical example under post-score and pre-exam submission. Section 4 outlines the general model, and in section 5 and 6 I generalize the results from example and include the discussion on pre-score submission, and Section 7 concludes. Omitted proofs can be found in Appendix.

2 Related Literature

This paper broadly connects three strands of literature. A first strand, initialized by Avery and Levin (2010), studies both students' application behavior and colleges' admission strategies in the context of decentralized college admissions. The tradeoff students face in the game induced by the Boston mechanism shares similarities with the decentralized game. For instance, Avery and Levin (2010) show when the early admission option is available in addition to the regular admission, students may apply early to a less popular college in hopes of a higher acceptance chance. The fact that centralized matching makes students application binding makes it closer to the early action than early decision. Chade, Lewis, and Smith (2014) examine a situation where students' application portfolios are constrained by the application cost, and colleges have uncertainties about students' ability. Che and Koh (2016) offer an theoretical analysis on yield management when students have uncertain preferences.

The second strand, started by Balinski and Sönmez (1999) and Abdulkadiroğlu and Sönmez (2003) emphasizes the importance of using mechanisms to improve the design of centralized college and school matching. Existing research in centralized choices is grounded on complete information with exceptions of Roth (1989), Chakraborty, Citanna, and Ostrovsky (2010) and Ehlers and Massó (2015) who mainly concern with strategy-proofness and stability. Abdulkadiroğlu, Che, and Yasuda (2011) and Miralles (2009) re-examined the potential ex-ante efficiency of the Boston mechanism, and demonstrate that Boston ex-ante dominates DA in situations with common ordinal preferences and no priorities. Troyan (2012) and Akyol (2014) relax assumptions on priorities and preferences respectively and provide criteria under which ex-ante efficiency of the Boston mechanism reemerges.

This paper follows the setting in the literature where students have common ordinal preference but different cardinal intensities, and consider a simpler priority structure where students are ranked the same at different colleges. The main departure from the previous papers on the ex-ante efficiency is that, instead of comparing Boston versus DA, I investigate different timings under the Boston mechanism. Since students may have any realization of scores under pre-exam submissions, and post-score submissions generates an outcome at equilibrium the same as the stable matching found by DA (Ergin and Sönmez, 2006), ex-ante efficiency is retained. In addition, ex-ante efficiency is reserved when signal is introduced under pre-score submissions.

The uncertainty clouded in the application process is endogenously entailed by the timing of preference submissions, which is an under-explored design dimension. A few recent papers study the use of timing under the Boston mechanism but with different focuses. Lien, Zheng, and Zhong (2016) consider a setting where exam scores may not reflect the true ability of students, and find that the timing with less information about scores is ex-ante fair as the students with higher ability is more likely to stay truthful at equilibrium compared with the student with lower ability. Pan (2017) takes a further step to understand behavioral aspect associated with timing, and disentangle overconfidence from ex-ante fairness using experiments.

Lastly, the welfare analysis on colleges relates to studies on sorting. There are mixed views with respect to the desirability of sorting (Pop-Eleches and Urquiola, 2013; Abdulkadiroğlu, Angrist, and Pathak, 2014). Instead of evaluating sorting, the main goal of this paper is to provide an positive analysis on the effect of timing on sorting. Calsamiglia, Martínez-Mora, Miralles, et al. (2015) study theoretically how Boston and DA affect socioeconomic sorting across ex-ante identical schools and neighborhoods. With residential priority, under both mechanisms students sort fully into quality-ranked schools and neighborhoods; Without residential priority though, only Boston generates sorting if there is a bad school. The presence of a bad school engages students in a equilibrium tradeoff to choose between a safe school versus their true choices. In another related paper, Wu and Zhong (2014) find field evidence that a top Chinese college is more likely to admit students with higher exam scores under post-score Boston and with lower exam scores under pre-exam Boston, lending empirical support to the predictions of the current model.

3 An Example

This section illustrates the main results of the paper using an example. There is a continuum mass of students to be assigned to three colleges, c_1, c_2, and c_3. Each

college has a capacity of 1/3.

Preferences. Students have common ordinal preferences: c_1, c_2, c_3 (that is, c_1 is the best college, and it is preferred over c_2, which is preferred over c_3). The underlying cardinal intensities vary: students receive 1 when attending c_1, $v \in [0, 1]$ when attending c_2, and zero from attending c_3. The student-specific parameter v measures preference intensity for c_2, or how enthusiastic the student feels about it. A higher value for v means the student finds c_2 almost as good as c_1. Preference intensity v is uniformly distributed on $[0, 1]$.

Priorities. Students receive a score e from the standardized exam which determines their priorities at colleges. A higher score is always preferred over a lower score. The priority order is strict and unique. Score e is uniformly distributed on $[0, 1]$. The distribution of v and e are independent, and they are common knowledge to students.

The Boston Mechanism. The Boston mechanism offers immediate acceptance, and it works as follows in a continuum setting:

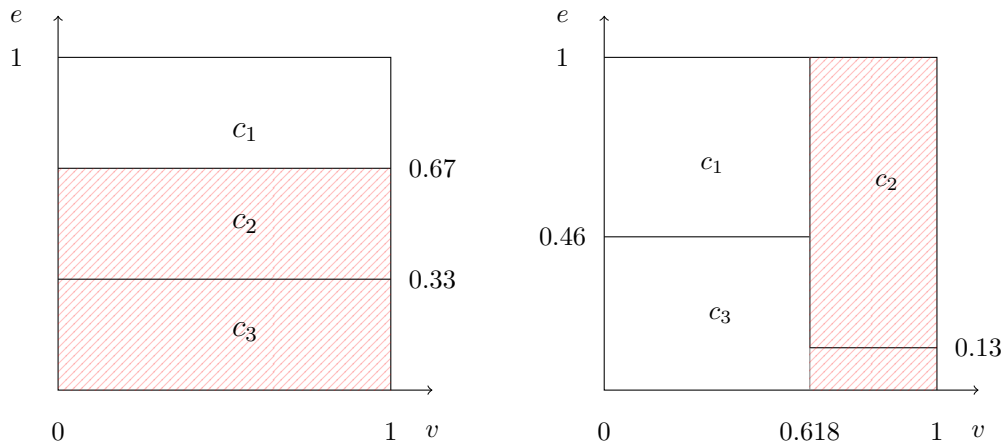
- In round 1, the first choices of the students are considered. Each college accepts the students up to its available capacity, such that the accepted ones have a score above the rejected ones, and reduces its capacity by the mass of students accepted.
- In round $r > 1$, the r-th choices of the students who were rejected in the previous round are considered. For each college that still has capacity, accepts the students by scores up to capacity, and reduces its capacity accordingly.
- The mechanism terminates when capacities are exhausted or there are no more students to be assigned.

Post-score submissions. Students know their scores when they submitting preferences under the post-score Boston mechanism.

First note that all students submitting preferences truthfully is not an equilibrium under post-score submissions. To see this, suppose they did submit truthfully, that is, all students submit a ROL as: $\pi_1 : c_1, c_2, c_3$. In the first round, c_1, the best college, admits those with a score above 2/3 (≈ 0.67), and its capacity is filled. No one ranks c_2 first in this round, and c_2 reserves full capacity for the next round. In the second round, students rejected by c_1 now apply to c_2, and only those with a score above 1/3 (≈ 0.33) will be accepted. The rejected students will apply in the next round to c_3 and all of them will be accepted.

Take a student who eventually goes to c_3, and consider the strategy where an alternative ROL $\pi_2 : c_2, c_1, c_3$ is submitted. Holding the behavior of all other students fixed, she will be accepted to c_2 in the first round, which yields her a higher

Figure 2: Equilibrium under the Boston mechanism



(A) Post-score submissions (B) Pre-exam submissions

Note: Students inside the white area submit preferences truthfully, whereas students inside the shaded region submit their preferences as $\pi_2 : c_2, c_1, c_3$. Equilibrium strategies result in the following matching: in Subfigure (A), students with a score no less than 0.67 are assigned to c_1, students with a score lower than that but no less than 0.33 are accepted at c_2, and the rest of students are assigned to c_3. In Subfigure (B), students who submit preferences truthfully and have a score no less than 0.46 are accepted by c_1, students who manipulate their preferences and have a score no less than 0.13 are accepted at c_2, and the rest of students are assigned to c_3.

expected utility than submitting preferences truthfully. Therefore, for students who are rejected by c_1, submitting π_2 weakly dominates submitting preferences truthfully.

Indeed, the following strategy constitutes an equilibrium strategy: submit preference truthfully if the score is above the threshold of c_1, and top rank c_2 otherwise.[2]

Pre-exam submissions. Students face uncertainty about the realization of scores under the pre-exam Boston mechanism.

Everyone submitting preferences truthfully is not an equilibrium under pre-exam submissions neither. To see this, again imagine they did submit preferences truthfully. Consider a student with preference intensity $v = 1$. Her probability of being accepted at c_1 is $1/3$ (≈ 0.33) in ignorance of her future score, then her expected utility when submitting preferences truthfully is 0.33.[3] Suppose now she submits $\pi_2 : c_2, c_1, c_3$ instead. Since no one applies to c_2 in the first round, then c_2 has capacity to accommodate her for sure, which yields her an expected utility of 1. Therefore, given that everyone else submits preferences truthfully, submitting π_2 yields a higher expected utility than truth-telling for this student.

On the other hand, pick a student with preference intensity $v = 0$, and fix the

[2]There are other outcome-equivalent equilibrium strategies, for example students with score less than 0.33 can top rank c_3 instead.

[3]The capacities in this example are such that at equilibrium both colleges exhaust their seats after the first round. In general, it can happen that by submitting truthfully there is still a positive probability to be assigned to c_2 in the next round.

strategy of everyone else being truth-telling, then submitting truthfully gives her an expected utility of 0.33, greater than the expected utility of top ranking c_2, which is 0.

In other words, students with a higher v now prefer to manipulate their preferences, whereas students with a lower v are willing to submit preferences truthfully, even at the risk of landing in c_3. This observation provides intuition for the equilibrium structure. The following cutoff strategy constitutes a Bayesian Nash equilibrium: submit preferences truthfully if $v \leqslant 0.618$, and submitting π_2 otherwise.

To check this is indeed an equilibrium, first note that at equilibrium a measure of 0.618 students apply to c_1 and a measure of 0.382 students applying to c_2 in round 1. This results in the capacity at both c_1 and c_2 filling up by the end of this round. The rejected students apply to c_3 in the next round and are accepted. For a student who submits preference truthfully, the probability of getting assigned to c_1 equals $0.33/0.618 \approx 0.539$, giving her an expected utility of 0.54. Likewise, the expected utility from submitting π_2 is $0.873v$. When the two expected utilities equalized, that is $v \approx 0.618$, students are indifferent between the two strategies.

Comparing timings. When comparing the two timings, there are a few features that arise as a consequence of the equilibrium strategies discussed above, and that can be extended to a more general model.

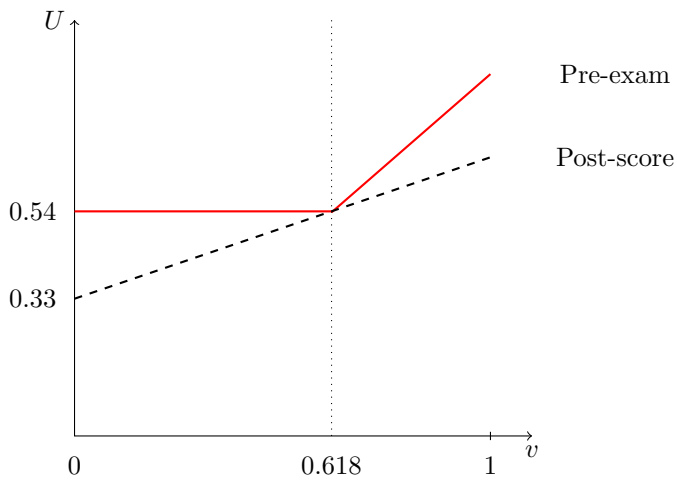
First, from an ex-ante perspective, students are better off under pre-exam submissions than post-score submissions.

Consider post-score submissions. For a given student with preference intensity v, the probability of being accepted at c_1 by submitting preferences truthfully is 0.33. Since one can only be accepted at c_2 by top-ranking it, the probability of being accepted at c_2 is thus the probability to be among the top 1/3 students among those would top-rank c_2 at equilibrium, that is, $(1/3)/(2/3) = 0.5$. If rejected, she will be assigned to c_3 for sure. Thus, her expected utility under post-score submissions is: $0.33 + 0.67 \times 0.5v = 0.33 + 0.33v$.

Next, consider pre-exam submissions. For a student who has weak preference intensity for c_2 and submits preference truthfully at equilibrium, with probability 0.54 she is accepted at c_1, and with probability 0.46 she is turned down and will be assigned to her least favorite choice c_3 in the end. Thus, her expected utility from truth-telling is 0.54. For a student with strong preference intensity for c_2 and top-ranks it at equilibrium, with probability 0.87 she is accepted at c_2, and with probability 0.13 she goes to c_3. This gives her an expected utility of $0.87v$. Thus, as Figure 3 depicts, for each value of preference intensity, pre-exam submissions dominate the post-score submissions.

Let us now turn our attention to colleges. For ease of exposition, I focus on c_1

Figure 3: Expected utility under the Boston mechanism



and c_2, as the acceptance cutoff remains unchanged for c_3.

Under post-score submissions, c_1 accepts students with a minimum score of 0.67, and c_2 accepts students with a score between 0.33 and 0.67. Thus, students with the best scores are accepted at the best college, and students with less good scores are accepted at the least good college, triggering full sorting on academic performance between the two colleges. When pre-exam submissions are in place, full sorting fails as some high scoring students are playing safe and are admitted now to c_2.

When taking a closer look at acceptance cutoffs, somewhat surprisingly, the fall of c_1's cutoff does not translate into a rise in c_2's cutoff. In fact, whenever c_2 has a capacity no less than c_1, its acceptance cutoff always go down as compared to post-score submissions. This is because colleges are forced to commit to its capacity each round in centralized admission system. Therefore, although c_2 is able to attract some high scoring students, it also has to admit low score ones in order to fill its capacity. This is a key difference from the decentralized college admission where colleges do not need to commit to fill their capacities in early admission round, and they can set their own admission standards to maintain the quality of incoming students (Avery and Levin, 2010).

To sum up, the example shows in a setting where students share common ordinal preferences but their cardinal intensities my differ, each student is weakly better off under pre-exam submissions than under post-score submissions. For colleges, post-score submissions triggers full sorting on academic performance, whereas sorting is only partial under the pre-exam submissions. However, as will be shown in the more general model, students' quality measured by average score is ambiguous under these two timings.

4 The model

The main ingredients in the model remain the same as in the example: there is a continuum mass of students with common ordinal preferences but different cardinal intensities to be matched a set of three colleges $\mathcal{C} = \{c_1, c_2, c_3\}$. Each college has a capacity of q_c. To mimic the tension that students who fail to attend the most popular college under the Boston mechanism often fail to go to the less popular college and land in a least preferred college, let $q_1 + q_2 < 1$. In addition, $\sum_{c=1}^{3} q_c = 1$, so that each student can be assigned to a college, avoiding outside option or waiting list.

Students' preferences are strict and all colleges are acceptable. For each student, let v_c denote the von Neumann-Morgenstern (vNM) utility value from attending college c. Without loss of generality, I normalize the vNM utility value for c_1 and c_3 to be 1 and 0 respectively, and let v be the vNM utility of attending c_2, which is drawn from $[0, 1]$ according to a cumulative distribution function G with density g.

Students' priorities are also strict and they are determined by exam score e, which is drawn on $[0, 1]$ according to a cumulative distribution function F with density f. Preference intensity and score are *independently* distributed, and both distributions are continuous, strictly increasing, and twice differentiable.

The signal x students receive under pre-score submissions is drawn according to a cumulative distribution function $H(x|e)$ on $[0, 1]$ with density function $f(x|e)$. The corresponding posterior of the signal has density function $f(e|x)$ on $[0, 1]$, with cumulative density function $F(e|x)$. Following the information literature, the signal satisfies strict *monotone likelihood ratio property* (MLRP) and *convex distribution function condition* (CDFC) (Milgrom, 1981; Grossman and Hart, 1983). By these two conditions, the acceptance chance is increasing and concave in signal. Finally, to ensure that a student with an exceptionally good signal should be almost never rejected, and that a student with an extremely bad signal should be almost always rejected, the following limit conditions are introduced: for a given e, $\lim_{x\to 1} F(e|x) = 0$, and $\lim_{x\to 0} F(e|x) = 1$.[4]

A *matching* in the continuum environment is a measurable allocation of students to colleges. To ease notations, I refer to θ as a student's type, where $\theta \equiv (v, e)$ under post-score and pre-exam submissions, and $\theta \equiv (v, x)$ under pre-score submissions. The allocation a student with type θ receives is $\mu(\theta)$. Then, a matching satisfies $\int \mathbb{1}_{\{\mu(\theta)=c\}} \, \mathrm{d}\eta(\theta) \leqslant q_c$ for all colleges, where η denotes the joint distribution for θ.[5]

[4]Exponential distribution is a classical example of distribution that satisfies the above assumptions, see Chade, Lewis, and Smith (2014) for such a case.

[5]The joint distribution of η is the product of G and F under post-score and pre-exam submissions, or the product of G and H under pre-score submissions, due to independences of these distributions.

Each student submits a rank-ordered list (ROL) over colleges denoted by π, and Π is the set of all submitted preference profiles. For each student, a strategy is a measurable mapping $\sigma : \theta \to \Pi$. A strategy profile is the set of strategies for all students. The solution concepts are Nash equilibrium for post-score submissions, and Bayesian Nash equilibrium for pre-exam and pre-score submissions. I search for symmetric pure strategy profiles as the existence of an equilibrium in continuum is guaranteed (Mas-Colell, 1984).

Throughout the paper, I consider the game induced by the Boston mechanism in continuum. Let $\pi(r)$ be the college in the r-th position of π, then the mass of students who demand for c in round r is

$$m_{c,r} = \int\int \mathbb{1}_{\{\theta:\pi(r)=c\}} \, \mathrm{d}\eta(\theta), \tag{1}$$

and the proportion of students who are applying to c with a minimum score of e_c is

$$a_{c,r}(e_c) = \int_{e_c}^{1} \int_{0}^{1} \mathbb{1}_{\{\theta:\pi(r)=c\}} \, \mathrm{d}\eta(\theta). \tag{2}$$

In the first round, students are considered by their first choices. The acceptance cutoff equals 0 if a college has more capacity available than the mass of students who demand it, and it is a positive value between 0 and 1 otherwise:

$$\hat{e}_{c,1} = \begin{cases} 0 & \text{if } q_c \geqslant m_{c,1}, \\ e \in (0,1) | a_{c,1}(e_c) = q_c & \text{otherwise .} \end{cases} \tag{3}$$

Accordingly, the capacity at each college is reduced by the amount of students accepted.

In the r-th round, students rejected from the previous round are considered by their r-th choices, and the acceptance cutoff for each college becomes:

$$\hat{e}_{c,r} = \begin{cases} 0 & \text{if } q_j - \sum_{j=1}^{r-1} m_{c,j} \geqslant m_{c,r}, \\ e \in (0,1] | a_{c,r}(e_c) = q_c - \sum_{j=1}^{r-1} m_{c,j} & \text{otherwise .} \end{cases} \tag{4}$$

The capacity at each college is further reduced by the amount of students accepted.

The final acceptance cutoff $\hat{e}_c$ equals the cutoff in the ending round.

5 Students' application strategies

5.1 Equilibrium with post-score submissions

As seen in the example, complete information about scores allows students to know precisely which college they will attend at the time of application, score is thus the only driving factor for equilibrium strategy. Since all colleges are acceptable and one can always be accepted at c_3, then I will focus on the two non-trivial ROLs that maximize the acceptance chance to one of the more popular colleges for each student: $\pi_1 : c_1, c_2, c_3$ and $\pi_2 : c_2, c_1, c_3$.

Proposition 1 (Post-score strategy)**.** *Consider the revelation game induced by the Boston mechanism under post-score submissions. The following strategy profile constitutes a Nash equilibrium: submit preferences truthfully if the student has a score above the acceptance cutoff of c_1, where $\hat{e}_1 = F^{-1}(1 - q_1)$, and submit $\pi_2 : c_2, c_1, c_3$ otherwise.*

Proposition 1 outlines one possible equilibrium. In this equilibrium, preference intensity does not have any impact on strategy even though it is allowed for the definition. A sketch of the proof follows. Consider students with score higher than the acceptance cutoff of c_1, submitting truthfully returns them higher payoff than submitting the other preference profile π_2. For students with lower score, submitting π_2 yields weakly higher payoff than submitting truthfully.

One caveat relates to the possibility of multiple equilibrium strategies. For example, the following strategy could also be an equilibrium: in addition to students who have a score higher than c_1's cutoff, those with a score lower than the acceptance cutoff of c_2 also submit preferences truthfully, while the rest of students top rank c_2. Despite of multiple equilibria, the matching outcome is unique. Ergin and Sönmez (2006) show that under general preference and priority profiles, the Nash equilibrium outcomes induced by the Boston mechanism coincide the set of stable matchings induced by DA when information is complete. In the current model, the priority is common and strict, then the stable matching is unique, which coincides with the Nash equilibrium outcome induced by the Boston mechanism.

5.2 Equilibrium with pre-exam submissions

I now analyze the equilibrium strategy of students under pre-exam submissions, where they have no information about the future realization of scores. As is evident in the example, preference intensity is the driving force for the equilibrium strategy under pre-exam submissions.

Although the precise acceptance cutoffs at each college are no longer known, the following observations about equilibrium cutoffs can be easily made.

Lemma 1. *In any equilibrium of the revelation game induced by the pre-exam Boston mechanism, the acceptance cutoffs satisfy: (1)* $\hat{e}_{c,r} \leqslant \hat{e}_{c,r+1}$*, and (2)* $\hat{e}_{1,2} = 1$*.*

The first part of the lemma is a direct consequence of the immediate acceptance property by the Boston mechanism, which rewards students who top rank it with a lower acceptance cutoff in an earlier round, and motivate them to be strategic about their first choices.[6] The second part of the lemma is due to the fact that c_1 always fills capacity after the first round. Suppose if there remain extra seats in the second round, given that all students prefer c_1 over the rest, then there must be some students who are not playing equilibrium strategies. These students can benefit from a positive acceptance chance in the first round by submitting preferences truthfully.[7]

Next, consider a student who knows that under pre-exam submissions, acceptance cutoffs never fall in the next round, and c_1 has an acceptance cutoff equal to the full score in the second round such that no one will be admitted. Given the strategies of others fixed, then at equilibrium the student prefers to submit preferences truthfully if the expected utility of getting into c_1 and c_2 by submitting truthfully exceeds the expected utility of getting into c_2 by top ranking it:

$$P_1(\pi_1) + P_2(\pi_1)v \geqslant P_2(\pi_2)v + P_1(\pi_2), \tag{5}$$

where $P_c(\pi)$ denotes the probability of this student being accepted by c when submitting π.

The condition above suggests that the equilibrium strategy involves a cutoff form depending on v. To see this, let ϕ be the expected payoff difference between submitting the two ROLs:

$$\phi \equiv P_1(\pi_1) + P_2(\pi_1)v - P_2(\pi_2)v - p_1(\pi_2). \tag{6}$$

Lemma 1 implies $P_1(\pi_2) = 0$ and $P_2(\pi_2) > P_2(\pi_1)$. Then the payoff difference falls in v as $\phi_v = P_2(\pi_1) - P_2(\pi_2) < 0$. Given that ϕ is monotonically decreasing and continuous in v, then if a student with taste v submits her preferences truthfully, another student with the preference intensity $v' < v$ also submits preferences truthfully, because the expected payoff of submitting preference truthfully relative to manipulating preference is decreasing in v. Vice versa, if a student with taste v manipulates her preference, then another student with taste $v' > v$ also manipulates.

[6] This is also true under the post-score submissions. However, since students observe their own scores, the result becomes trivial.

[7] The example in Section 3 is a special case that the acceptance cutoffs of both c_1 and c_2 are equal to 1 in the second round.

Once students' equilibrium strategies are known to take a cutoff form, it is easy to express the probability of acceptance $P_c(\pi)$ in terms of the equilibrium cutoff value $\hat{v}$, and use $\phi = 0$ to pin down the value of $\hat{v}$. The next proposition is proved in Appendix A.1.

Proposition 2 (Pre-exam strategy)**.** *Consider the revelation game induced by the pre-exam Boston mechanism. The following strategy profile constitutes a Bayesian Nash equilibrium: submit preferences truthfully if $v \leqslant \hat{v}$, and submit $\pi_2 : c_2, c_1, c_3$ otherwise, where $\hat{v}$ is uniquely pinned down by*

$$\frac{q_1}{G(\hat{v})} + \max\left\{0, \frac{q_2 - 1 + G(\hat{v})}{G(\hat{v}) - q_1}\right\}\hat{v} = \min\left\{1, \frac{q_2}{1 - G(\hat{v})}\right\}\hat{v}, \tag{7}$$

and $G(\hat{v})$ is the mass of students who submit preferences truthfully at equilibrium.

In this cutoff equilibrium, those with a strong preference intensity for c_1 submit preferences truthfully, despite that they may achieve low scores and as a result assigned to the worst option. On the other hand, those with a relatively strong preference intensity for c_2 top rank it instead, forfeiting the possibility of obtaining high scores in the future. A closed form solution of the equilibrium cutoff $\hat{v}$ requires to specify the distribution of preference intensity. Corollary 1 offers some comparative statics of $\hat{v}$ in relation to capacities.

Corollary 1. *The cutoff value for preference intensity $\hat{v}\,(q_1, q_2)$ increases in q_1, and first decreases then increases in q_2.*

The intensity cutoff value $\hat{v}$ is monotone in q_1, but not in q_2. Therefore, increasing the capacity of c_1 makes truth-telling more attractive. Whereas for c_2, q_2 influences both the probability of getting into it in the first round, $P_2(\pi_2)$, and in the second round, $P_2(\pi_1)$. Then increasing q_2 attracts more students to top rank c_2 initially, but when the capacity becomes too abundant, it reduces the attractiveness of top ranking c_2 relative to truth-telling.

5.3 Equilibrium with pre-score submissions

I now discuss the intermediate case where instead of having full or no information about scores, students receive a signal that allows them to infer future score.[8] As soon will be clear, both signal and preference intensity drive the equilibrium strategy.

[8]This timing resembles the situation where the pre-exam submissions are used and students acquire additional information about the realization of score in the future, for example, through past test grades.

Let $P_c(\pi, x)$ denote the probability of getting into college c by submitting π when receiving signal x. Given others' strategies fixed, at equilibrium the student prefers to submit preferences truthfully if:

$$P_1(\pi_1, x) + P_2(\pi_1, x)v \geqslant P_2(\pi_2, x)v - P_1\pi_2, x. \tag{8}$$

Lemma 1 remains valid under pre-score submissions, then the equilibrium strategy involves a cutoff strategy on preference intensity v for each signal x. The next proposition is proved in Appendix A.3.

Proposition 3 (Pre-score strategy)**.** *Consider the revelation game induced by the Boston mechanism under pre-score submissions, the following strategy constitutes a Bayesian Nash equilibrium: submit preferences truthfully if $v \leqslant \hat{v}(x)$, and submit $\pi_2 : c_2, c_1, c_3$ otherwise. The cutoff $\hat{v}(x)$ rises in x, and for a given x, $\hat{v}(x)$ is uniquely given by*

$$\frac{q_1}{\tilde{m}} + \max\left\{0, \frac{q_2 - 1 + \tilde{m}}{\tilde{m} - q_1}\right\}\hat{v}(x) = \min\left\{1, \frac{q_2}{1 - \tilde{m}}\right\}\hat{v}(x), \tag{9}$$

where $\tilde{m} \equiv \int_0^1 G(\hat{v}(x))h(x|e)\,\mathrm{d}x$ is the mass of students who submit preferences truthfully at equilibrium.

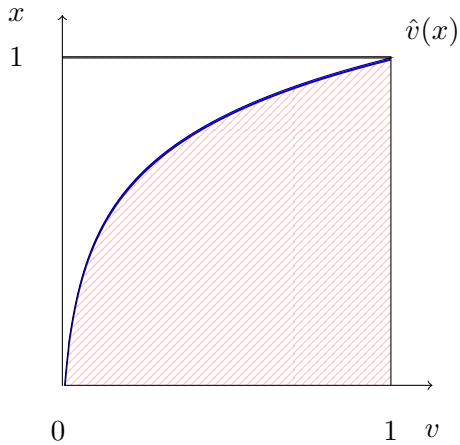
A new feature of the equilibrium compared to pre-exam submissions is that the cutoff strategy relies on signal x in addition to preference intensity v (see Figure 4 for an illustration).

The cutoff function $\hat{v}(x)$ has a few properties worth discussing. First, $\hat{v}(x)$ is increasing in signal x. In general, students who receive worse signal have lower chance to obtain good score, and thus they prefer top-ranking the alternative option c_2. On the other hand, better signal increases the acceptance chance to c_1 relative to c_2, and thus they prefer submitting preferences truthfully.

Second, $\hat{v}(x)$ is convex, meaning an incremental quality in signal convinces more students to submit preferences truthfully when signal about score is high than low. This is because the additional gain in acceptance chance to c_1 brought by an increment of signal relative to c_2 increases as implied by CDFC of the signal.

Moreover, $\hat{v}(x)$ is closer to the 45-degree diagonal when the overall quality of signal becomes noisier, and is closer to $(0, 1)$ when quality of signal becomes finer. When signal becomes completely informative, the cutoff locus jumps to the acceptance cutoff under the post-score submissions; When signal is merely babbling, the cutoff locus becomes the vertical line in the pre-exam submissions.

Figure 4: Equilibrium under the pre-score Boston mechanism



Note: The white area represents students submit $\pi_1 : c_1, c_2, c_3$, and the shaded area corresponds to students submitting $\pi_2 : c_2, c_1, c_3$.

6 Comparing timings

6.1 Ex-ante welfare for students

This subsection compares students' welfare from an ex-ante perspective, that is, I investigate students' expected utility before they learn about their scores or signals under each timing, given the equilibrium strategy described in Section 5.

Proposition 4. *Under the Boston mechanism, students are weakly better off under pre-exam submissions, than under pre-score submissions, and than under post-score submissions.*

The proposition suggests that students obtain higher ex-ante welfare when they have less information about scores. The less information that students have, the more likely equilibrium strategy is driven by preference intensity, leading to a higher welfare for everyone. For intuition, consider first the case of post-score submissions. Preference intensity is ignored at equilibrium, resulting in the lowest welfare for everyone. This situation is improved when less information about scores is available. Acceptance chance to c_1 increases for those with stronger preference for it, while acceptance chance to c_2 does not decrease for those with less disutility for it.

In addition, expected utility is higher under pre-exam than pre-score submissions. Signal reduces the desirability of applying purely based on cardinal intensity. As a result, students with higher signal top rank c_1 despite less enthusiasm, while students with medium signal top rank c_2 despite less enthusiasm, intensifying competition at both colleges as compared to the case of pre-exam submissions.

The ex-ante welfare comparison relates to the welfare result of (Abdulkadiroğlu, Che, and Yasuda, 2011) where a lottery is used to break ties in school choice. In the

current setting, students are considered totally ignorant of their scores under pre-exam submissions, this is as if their future score is determined by a lottery according to some distribution. The results here further suggest that the ex-ante dominance over post-score submissions carries through if students acquire some information about their scores.

6.2 Sorting at colleges

Colleges in the centralized admission system have limited power to influence who to admit once mechanism is chosen.[9] Will best students always apply to best colleges when timing of preference submissions changes? Which timing colleges prefer if they only care about the academic quality of incoming students?

To answer these questions, I first look at acceptance cutoffs under three timings. The following lemma is proved in Appendix A.5.

Lemma 2. *(1) Under post-score submissions, $\hat{e}_1^1 > \hat{e}_2^1$; Under pre-exam and pre-score submissions, $\hat{e}_1^t > \hat{e}_2^t$ if c_2 is larger than c_1. (2) For c_1, $\hat{e}_1^1 > \hat{e}_1^3 > \hat{e}_1^2$; For c_2, $\hat{e}_2^3 > \hat{e}_2^1 > \hat{e}_2^2$.*

It is not surprising that c_1 always maintains a higher admission cutoffs than c_2 under post-score submissions. This is also the case if c_1's capacity is less than that of c_2. Across all timings, c_1 enjoys the highest acceptance cutoff under post-score submissions. However, it is less straightforward how c_2's acceptance cutoff changes in response to the timing. Under pre-exam submissions, despite that c_2 can attract some of the high scoring students, it also loses students with medium scores who it could have accepted under post-score submissions, which causes the fall of its admission cutoffs. On the other hand, when students receive a signal under pre-score submissions, although c_2 can attract a smaller fraction of high scoring students, it keeps more medium score students, resulting in a even higher acceptance cutoff than under post-score submissions.

With the relation of acceptance cutoffs in mind, I can now compare the quality of students between the two colleges for each timing, and the quality of students for each college across three timings. For the between-college comparison, I follow the definitions on sorting first introduced by Calsamiglia, Martinez-Mora, and Miralles (2014). Let ϕ_c^t denote the cumulative distribution function of students' scores *conditional on being accepted into* c under timing t, and $supp(\Phi_c^t)$ denote the support of Φ_c^t. There is **full sorting** between two more popular colleges if $\inf(supp(\Phi_{c_1}^t)) \geqslant \sup(supp(\Phi_{c_2}^t))$, and there is **partial sorting** between the two

[9] In contrast, colleges in the decentralized admission system can actively screen students for example by setting different admission standards for early applicants and regular applicants (**?**).

more popular colleges if $\Phi^t_{c_1}$ first-order stochastically dominates (FOSD) $\Phi^t_{c_2}$. For the across-timing comparison, I use average score of incoming students to measure quality of students.

Proposition 5. *Under Boston mechanism:*
(1) Post-score submissions generate full sorting between the two colleges, whereas the pre-exam and pre-score submissions generates partial sorting;
(2) In terms of average score of students, c_1 prefers post-score to pre-score to pre-exam submissions, and c_2 prefers pre-score to both post-score and pre-exam submissions.

The results suggest first, the validity of sorting depends on the timing in place. Under post-score submissions, best students always go to c_1, and less good students go to c_2. Full sorting fails both under pre-score and pre-score submissions. Second, pre-score submissions allow c_2 to improve its students' quality as compared to pre-exam and post-score submissions. However, comparing students' composition for c_2 under pre-exam and post-score submissions is more subtle. Indeed, c_2 receives both higher and lower score students under pre-exam submissions, and the distribution of scores admitted to c_2 under pre-exam submissions is second-order stochastically dominated by the distribution of scores admitted to it under post-score submissions.

7 Concluding remarks

I have formulated the game induced by the Boston mechanism in order to study the effects of timing to preference submissions. Students have common ordinal preferences but different cardinal preference intensities, and colleges rank students according to a common priority structure determined by exam scores. Timing affects students to adopt different equilibrium strategies, which in turn lead to two welfare effects. First, timing where less information about score is provided gives higher expected payoff for students. Second, full sorting is reduced when post-score submissions changes to to pre-score or pre-exam submissions.

The effects highlighted in the paper are probably not the only reasons why clearinghouses adopt such timings. One alternative explanation could be the logistic concerns. For instance, submitting preferences after publication of scores requires coordination in a short time in order to run the match, whereas submit preferences beforehand could reduce the administrative load for clearinghouses.

While three colleges and homogeneous ordinal preferences are restrictive, the model provides a parsimonious setting to analyze equilibrium strategies across timings. One can think of the current model in a setting with more than three colleges. The generalization of equilibrium strategies under post-score submissions is

straightforward, whereas under pre-exam and pre-score submissions more equilibrium conditions are needed. Moreover, students preferences may be heterogenous. The discussion on this point can be found in Appendix A.7. If some students prefer the less popular as their first choice, either due to closer location or better financial aid, then these students still have higher expected payoff under pre-exam and pre-score submissions than the post-score submission, whereas for those students who still prefer the more popular college as their top choices, not everyone benefits from less information. This is because the presence of heterogeneity decreases incentive to manipulate preferences at equilibrium, leading into less efficiency gain. On the other hand, full sorting does not exist anymore in the case of heterogenous preferences.

Finally, this paper aims to provide a positive analysis on the use of timing. The results suggest that when taking ex-ante efficiency or sorting into account, pre-exam or pre-score submissions under the non-strategyproof mechanism such as the Boston mechanism can be useful. Of course, the Boston mechanism is not the only non-strategyproof mechanism, generalization to other non-strategyproof mechanisms invites further investigation.

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A Omitted Proofs

A.1 Proof of Proposition 2

I first show the cutoff $\hat{v}$ exists. Note that the probability of being accepted in college c submitting π where the submitted ROL $\pi(r) = c$ equals the probability that the student has a score higher than the cutoff of c when her application is considered in round r,

$$P_c(\pi) = 1 - F(\hat{e}_{c,r}) = \min\left\{1, \frac{q_c - \sum_{j=1}^{r-1} m_{c,j}}{m_{c,r}}\right\}. \tag{A.1}$$

The remainder of the proof compromises two cases. Consider first that capacity at c_2 is also filled after the first round, that is, $P_2(\pi_1) = 0$. The measure of students who compete for college c_1 and c_2 in the first round are $m_{1,1} =$

$\int\int \mathbb{1}_{\{(v,e):\pi_1(1)=c_1\}}\,\mathrm{d}G(v)\,\mathrm{d}F(e) = G(\hat{v})$, and $m_{2,1} = 1 - G(\hat{v})$ respectively. Then, $P_1(\pi_1) = \frac{q_1}{G(\hat{v})}$, and $P_2(\pi_2) = \frac{q_2}{1-G(\hat{v})}$. Evaluating Equation (5) at zero yields:

$$\frac{q_1}{q_2} = \frac{\hat{v}G(\hat{v})}{1 - G(\hat{v})} \equiv \gamma(\hat{v}). \tag{A.2}$$

It is easy to show that $\gamma_{\hat{v}} \geqslant 0$, implying $\gamma(\hat{v})$ is non-decreasing and crosses $\hat{v}$ once from above. Then, there must exist a fixed point such that $\gamma(\hat{v}) = \hat{v}$, and the solution to equation (A.2) exists.

In the second case, there is excess capacity at c_2 in the second round. Then students who submit π_2 are accepted with probability 1 to c_2 in the first round. In addition, $P_2(\pi_1) = q_2 - \frac{1-G(\hat{v})}{G(\hat{v})-q_1}$, and $P_1(\pi_1) = \frac{q_1}{G(\hat{v})}$. Evaluating Equation (5) again at zero returns:

$$\frac{q_1}{1 - q_1 - q_2} = \frac{\hat{v}G(\hat{v})}{G(\hat{v}) - q_1} \equiv \gamma(\hat{v}). \tag{A.3}$$

Now $\gamma_{\hat{v}} \leqslant 0$, then $\gamma(\hat{v})$ is non-increasing and crosses $\hat{v}$ once from below. Therefore, there also exists a fixed point such that $\gamma(\hat{v}) = \hat{v}$, and the solution to equation (A.3) also exists.

Uniqueness of $\hat{v}$ is guaranteed by continuity. Suppose there exits another cutoff equilibrium $\hat{v}'$ such that students top rank c_2 if her preference intensity is higher than this cutoff, and top rank c_1 otherwise. Since v is continuous, then for some students, it contradicts the fact that $\hat{v}$ is also a cutoff at equilibrium.

A.2 Proof of Corollary 1

Evaluating Equation (5) at zero, then the payoff difference from submitting π_1 versus π_2 at equilibrium is:

$$\phi = \begin{cases} \dfrac{q_1}{G(\hat{v})} - \dfrac{q_2\hat{v}}{1 - G(\hat{v})}, & \text{if } q_2 \leqslant 1 - G(\hat{v}); \\ \dfrac{q_1}{G(\hat{v})} - \dfrac{1 - q_1 - q_2}{G(\hat{v}) - q_1}\hat{v}, & \text{otherwise.} \end{cases}$$

Note that $\phi_{\hat{v}} < 0$. In the first case, $\phi_{q_1} > 0$, using implicit function theorem, then for a given q_2, $\hat{v}_{q_1} > 0$; Similarly, $\phi_{q_2} < 0$, then for a given q_1, $\hat{v}_{q_2} < 0$. In the second case, $\phi_{q_1} > 0$ still holds, but $\phi_{q_2} > 0$. Then, for a given q_2, $\hat{v}_{q_1} > 0$, and similarly for a given q_1, $\hat{v}_{q_2} > 0$.

A.3 Proof of Proposition 3

Suppose that for each signal x, the cutoff $\hat{v}(x)$ exists. The probability of being

accepted in college c by submitting π where $\pi(r) = c$ is:

$$P_c(\pi, x) = 1 - F(\hat{e}_{c,r}|x) = \min\left\{1, \frac{q_c - \sum_{j=1}^{r=1} \tilde{m}_{c,j}}{\tilde{m}_{c,j}}\right\}. \tag{A.4}$$

Compared to the pre-exam submissions, now the mass of students who will be considered by c is different in a given round. Let $\tilde{m}$ denote the mass of students who are truth-telling, i.e. the mass of students who demand for c_1 in round 1:

$$\tilde{m} = \int_0^1 \int_0^{\hat{v}(x)} g(v)h(x|e)\,\mathrm{d}v\,\mathrm{d}x = \int_0^1 G(\hat{v}(x))h(x|e)\,\mathrm{d}x, \tag{A.5}$$

while the measure of student top rank c_2 is $1 - \tilde{m}$.

The remainder of the proof again comprises two cases. First, when the capacity of c_2 is also exhausted after the first round, the following equilibrium condition can be obtained:

$$\frac{q_1}{q_2} = \frac{\hat{v}(x)\int_0^1 G(\hat{v}(x))h(x|e)\,\mathrm{d}x}{\int_0^1 (1 - G(\hat{v}(x)))h(x|e)\,\mathrm{d}x} \equiv \gamma(\hat{v}(x)). \tag{A.6}$$

Again, $\gamma_{\hat{v}} \geqslant 0$, suggesting that $\gamma(\hat{v}(x))$ weakly rises in $\hat{v}(x)$, and it crosses $\hat{v}(x)$ only once from above. Then, for a given x, there exists a fixed point such that $\gamma(\hat{v}(x)) = \hat{v}(x)$, confirming that the solution to Equation (A.6) exists. In the second case where c_2 still has capacity for the second round, one can also verify that, for a given x, preference cutoff $\hat{v}(x)$ exists. Moreover, continuity guarantees the uniqueness.

Finally, to see the properties of preference cutoff $\hat{v}(x)$, note that $1 - F(e|x)$ is strictly supermodular in (e, x) since the posterior density function $f(e|x)$ satisfies MLRP. In the first case where c_2 runs out capacity after the first round, I assume $\hat{e}_1 > \hat{e}_2$ without loss of generality. Then, the following term — the ratio of the probability of getting into c_1 relative to the probability of getting into c_2 — rises in x,

$$\frac{1 - F(\hat{e}_1|x)}{1 - F(\hat{e}_2|x)}.$$

This in turn suggests that $\hat{v}$ must also rises in x.

In the second case where c_2 still has capacity, since students rejected by c_1 are able to compete for the remaining seat in c_2 in the second round, then it follows that $\hat{e}_1 > \hat{e}_{2,2}$. Therefore, the following term — the probability of getting into c_1 by submitting π_1 relative to the difference between probabilities of getting c_2 by

submitting π_1 and π_2 — rises in x,

$$\frac{1 - F(\hat{e}_1|x)}{F(\hat{e}_{2,2}|x) - F(\hat{e}_{2,1}|x)}.$$

This concludes the proof that $\hat{v}(x)$ must increase in x.

A.4 Proof for Proposition 4

First of all, I introduce formally the definition of expected utility under different timings. The ex-ante expected utility under timing t is:

$$U^t = \sum_{c \in \mathcal{C}} P_c(\sigma^{t*}(\theta)) v_c,$$

where $P_c(\sigma^{t*}(\theta))$ denotes the probability of being assigned to c for a student with type θ.

Next, I show both pre-exam and pre-score Boston mechanism ex-ante dominates post-score Boston. I only present the pre-score case here as the proof for the case of pre-exam is essentially the same. The proof is done by two claims.

Claim 1. *Consider any symmetric equilibrium under the post-score Boston mechanism. The expected utility to any student given that other students follow the equilibrium strategy described by Proposition 1 is* $U^1 = \sum_{c \in \mathcal{C}} q_c v_c$.

Consider any symmetric equilibrium under the post-score Boston mechanism. The expected utility to any student given that other students follow the equilibrium strategy described by Proposition 1 is $U^1 = \sum_{c \in \mathcal{C}} q_c v_c$. To see this, note that under the post-score Boston mechanism, the ex-ante probability of getting assigned to a college c at equilibrium is

$$P_c(\sigma^{1*}(\theta)) = \left(\frac{1 - q_1}{1}\right)\left(\frac{1 - q_1 - q_2}{1 - q_1}\right) \dots \left(\frac{q_c}{1 - q_1 - \dots - q_{c-1}}\right) = q_c.^{10} \tag{A.7}$$

Claim 2. *Consider any symmetric equilibrium under the pre-score Boston mechanism. Then, suppose other students follow the equilibrium strategy described by Proposition 1, there exist a strategy for any student that gives the same expected utility as under the post-score Boston mechanism.*

[10] This probability is the same as the ex-ante probability of getting assigned to a college under DA. Under the post-score Boston mechanism, the equilibrium strategy is such that a student is assigned to the more popular college c_1 when she has a high enough score (and submit preferences truthfully), or she is assigned to the less popular college c_2 when her score fails to win her a seat at c_1 (and top rank c_2). In our current analysis, we only need to deal with 3 colleges, while the fact that the probability of getting into c under the post-score Boston mechanism holds more generally.

To see this, fix the strategies of others, a student with type θ adopts $\sigma^{3*}(\theta)$ at equilibrium. For each college $c \in \mathcal{C}$, the total mass of students assigned to each college equals the available capacity:

$$\int_0^1 P_c(\sigma^{3*}(\theta))h(x|e)\,\mathrm{d}x = q_c. \tag{A.8}$$

Fix a student with type $\tilde{\theta}$. Suppose this student, rather than playing the equilibrium strategy described in Proposition 2, picks a different strategy $\tilde{\sigma}^3 \equiv \sigma^{3*}(\theta)f(v)$, that is, she plays $\sigma^{3*}(\theta)$ with probability $f(v)$. This effectively implies that she ignores her preference intensity. Then, we have

$$P_c(\tilde{\sigma}^3) = \int_0^1 P_c(\sigma^{3*}(\theta))h(x|e)\,\mathrm{d}x = q_c = P_c(\sigma^{1*}(\theta)). \tag{A.9}$$

Since $\tilde{\sigma}^3$ is not equilibrium strategy, then we must have

$$\sum_{c\in\mathcal{C}} v_c P_c(\sigma^{3*}(\tilde{\theta})) \geqslant \sum_{c\in\mathcal{C}} v_c P_c(\tilde{\sigma}^3) = \sum_{c\in\mathcal{C}} v_c P_c \sigma^{1*}(\theta). \tag{A.10}$$

Lastly, to show that the pre-exam Boston mechanism ex-ante dominates the pre-score Boston mechanism is to show $P_c(\sigma^3 * (\theta)) \leqslant P_c(\sigma^2 * (\theta))$.

Consider the pre-score Boston mechanism. Compared to the pre-exam timing, c_1 attracts an additional of $\int_{\hat{x}(\hat{v})}^1 (1 - G(\hat{v}(x))h(x|e)\,\mathrm{d}x$ students who were previously ranking c_2 first, where $\hat{x}(\hat{v})$ is the signal that associates with $\hat{v}$ from the pre-exam timing. These new applicants always have higher scores than the amount of $\int_0^{\hat{x}(\hat{v})} G(\hat{v}(x)h(x|e)\,\mathrm{d}x$ students it loses. Therefore, the competition for c_1 becomes tougher, then $P_1(\sigma^{3*}(\theta)) < P_1(\sigma^{2*}(\theta))$. For c_2, the inequality relation holds as c_2 attracts more medium score student compete for c_2 as their first choices (implied by concavity of the cutoff function $\hat{v}(x)$), lowering the overall probability of getting assigned to c_2.

A.5 Proof of Lemma 2

(a) *Acceptance cutoffs between colleges.* Equilibrium strategy under post-score submissions guarantees that $\hat{e}_1^1 > \hat{e}_2^1$ for any q_1 and q_2.

For pre-exam and pre-score submissions, there are two cases. (1) Consider first the case where q_2 exhausts after the first round. Equilibrium condition from Equation 7 together with $q_2 \geqslant q_1$ imply that $G(\hat{v}) > 1/2$, that is, more than half of the students submit preferences truthfully at equilibrium. Then, $\hat{e}_1^2 > \hat{e}_2^2$ as more students are competing for c_1 for first choice while c_1 has less capacity. The same logic can be followed to show that the equilibrium condition $\hat{e}_1^3 > \hat{e}_2^3$. (2) The case where

c_2 is not filled after the first round is trivial. In the second round c_2 can only accept students among those who are previously rejected from c_1, resulting the acceptance cutoff in this round, which is also the final cutoff, lower than the acceptance cutoff of c_1.

(b) *Acceptance cutoffs across timings.* For c_1, post-score submissions allow it to admit all students with the highest scores up to its capacity regardless of their enthusiasm, whereas it looses some high-score students under pre-score submissions and all high-score students with low enthusiasm for it under pre-exam submissions. Thus, the acceptance cutoff across three timings are such that $\hat{e}_1^1 > \hat{e}_1^3 > \hat{e}_1^2$.

For c_2, I present the case where c_2 is filled after the first round, the intuition for the second case is similar. Recall the acceptance cutoffs under three timings are $\hat{e}_2^1 = F^{-1}(1 - q_1 - q_2)$, $\hat{e}_2^2 = F^{-1}\left(1 - \frac{q_2}{1-G(\hat{v})}\right)$, and $\hat{e}_2^3 = F^{-1}\left(1 - \frac{q_2}{1-\tilde{m}}\right)$ respectively.

To see $\hat{e}_2^1 > \hat{e}_2^2$, note that the pre-score equilibrium condition from Equation 7 suggests that applying to c_2 has a higher acceptance chance than applying to c_1. Thus, $\frac{q_2}{1-G(\hat{v})} > \frac{q_1}{G(\hat{v})}$. This implies $\frac{q_2}{1-G(\hat{v})} > q_1 + q_2$, and thus $\hat{e}_2^1 > \hat{e}_2^2$.

The relation that $\hat{e}_2^3 > \hat{e}_2^1$ is due to the convexity of $\hat{v}(x)$ under pre-score submissions. The convexity of $\hat{v}(x)$ indicates that at equilibrium the mass of students who top rank c_2 is more than the mass of students who top rank c_1. Pre-score submissions allow c_2 to admit some of the students who have score above $\hat{e}_1^1$ that it could not have admitted under post-score submissions, thus for the same level of capacity, the acceptance threshold now rises.

A.6 Proof of Proposition 5

(a) *Sorting between colleges.* First, under post-score submissions, note that c_1 admits all students with a score in the range of $[\hat{e}_1^1, 1]$, and c_2 admits students with a score in the range of $[\hat{e}_2^1, \hat{e}_1^1]$, that is, the student with the lowest score admitted to c_1 has a score no lower than the student with the highest score admitted to c_2, generating full sorting. Second, under pre-exam submissions, both c_1 and c_2 admit high score students, yet $\hat{e}_1^2 > \hat{e}_2^2$, then for every score, c_1 admits weakly more students than c_2, thus, $\Phi_{c_1}^2$ FOSD $\Phi_{c_2}^2$. Similarly, $\Phi_{c_1}^3$ FOSD $\Phi_{c_2}^3$.

(b) *Students' quality across timings.* For c_1, it admits students with score range $[\hat{e}_1^1, 1]$, $[\hat{e}_1^3, 1]$, and $[\hat{e}_1^2, 1]$ under three timings respectively. Let $E(c^t)$ be the average score of admitted students at college c under timing t. Then, $E(c_1^1) > E(c_1^3) > E(c_1^2)$, as $\hat{e}_1^1 > \hat{e}_3^1 > \hat{e}_2^1$. For c_2, it admits students with score range $[\hat{e}_2^1, \hat{e}_1^1]$, $[\hat{e}_2^3, 1]$, and $[\hat{e}_2^3, 1]$ under three timings respectively. Since $\hat{e}_2^3 > \hat{e}_2^1$, then $\Phi_{c_2}^3$ FOSD $\Phi_{c_2}^1$. For a non-decreasing utility value, FOSD implies that the expected value $E(c_2^3) > E(c_2^1)$. Similarly, $\hat{e}_2^3 > \hat{e}_2^2$, then $\Phi_{c_2}^3$ FOSD $\Phi_{c_2}^2$, and consequently $E(c_2^3) > E(c_2^2)$. The domain of $\Phi_{c_2}^2$ contains $\Phi_{c_2}^1$, that is c_2 admits both higher score students and lower

score students under pre-exam submissions as compared to post-score submission, then $\Phi^1_{c_2}$ SOSD $\Phi^2_{c_2}$, where SOSD represents second order stochastic dominance. Note in this case, FOSD fails, thus average score of students under these two timings depend on the choice of distribution.

A.7 Heterogeneous ordinal preferences

This section discusses the heterogeneous case in a fashion similar to the Hotelling model. When ordinal preferences are heterogeneous, a caveat is that ex-ante efficiency of the Boston mechanism under pre-exam or pre-score submissions disappear. The reduction of sorting effect persists. However, to start with, the problem of full sorting is less severe than the case where everyone's ordinal preferences are identical.

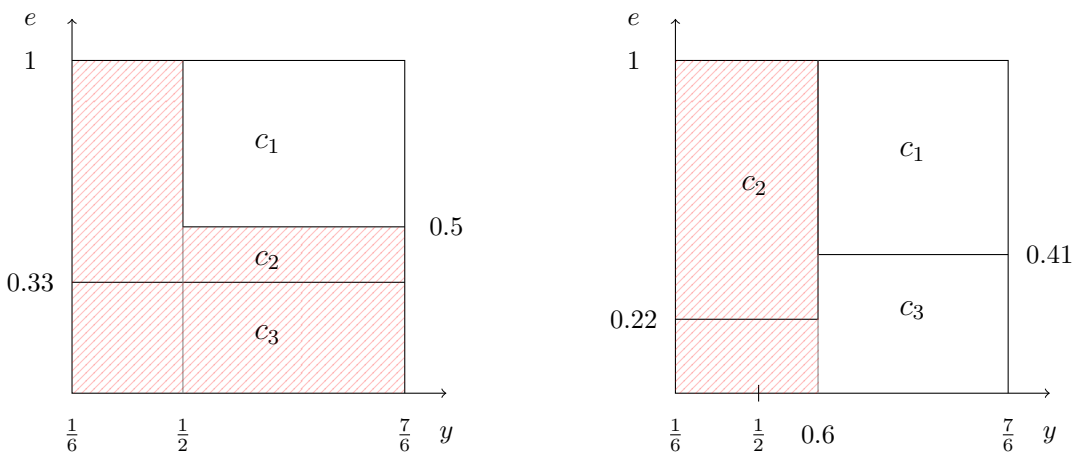
Again to keep the results stark, I focus on the case of three colleges.[11] Following (Coles, Kushnir, and Niederle, 2013), I consider students with block preferences: they all agree c_1 and c_2 are better than c_3, but disagree on whether c_1 or c_2 is the most desirable. In particular, a student's payoff is y when attending c_1, and $1 - y$ when attending c_2, while attending c_3 gives zero payoff. Thus, the interpretation of y changes as compared to preference intensity v of the main model. Now y measures the preference intensity for c_1 relative to c_2. The higher the value of y, the stronger a student's preference intensity for c_1. Therefore, students prefer c_1, c_2, c_3 if $y \geqslant 1/2$, and c_2, c_1, c_3 otherwise, where $y = 1/2$ a student is indifferent between the two better colleges. Let $G(y)$ denote the cumulative distribution function of the new preference intensity parameter y on an interval of $[a, 1 + a]$, where $a \in [0, 1/2]$. The distribution of preference intensity y and score e are independent.

I consider the more interesting case where c_1 is still the most popular college, that is, $G(1/2) < 1/2$. Equilibrium strategy is still monotone despite of heterogeneous ordinal preferences. Figure 5 illustrates the equilibrium strategy under post-score and pre-exam submissions. Comparing to the homogeneous case, at equilibrium the mass of students manipulating at equilibrium is smaller, as heterogeneity — some preferring c_2 as their first choice — reduces the acceptance chance to c_2 even by top ranking it. In addition, manipulation at equilibrium no longer improve ex-ante efficiency for everyone.

Indeed, as demonstrated in Figure 6, students with $y < 1/2$ prefer pre-exam submissions to post-score submissions, however the situation for students with $y \geqslant 1/2$ is ambiguous. The pre-score submissions favor those with either high preference intensity (and stay truthful at equilibrium), or those with low preference intensity

[11] For a generalized Hotelling model of n colleges, each college can be considered of situating in the vertex of the simplex (Somaini and Einav, 2013). When there are two good colleges, the simplex is simply a square, and when there are three good colleges, the simplex becomes a equilateral triangle.

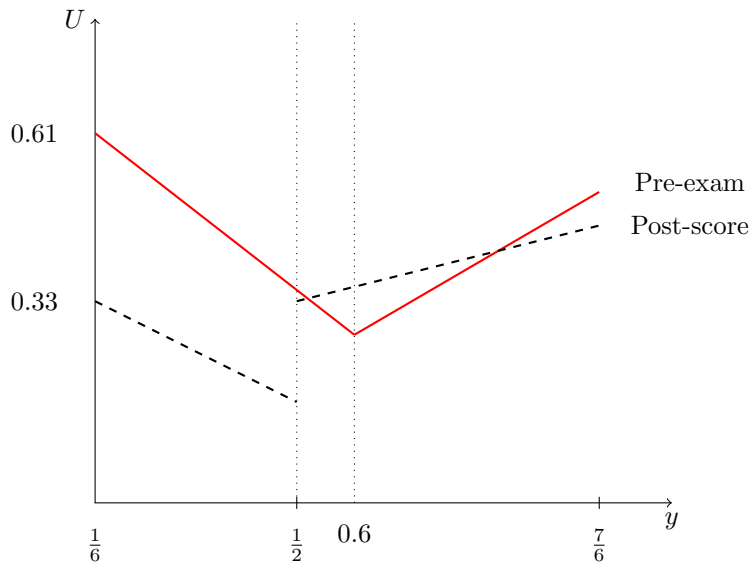
Figure 5: Application strategies under heterogeneous ordinal preferences



(A) Post-score submissions (B) Pre-exam submissions

Note: (1) The example uses $q_c = 1/3$. The score $e \sim U[0,1]$, and preference intensity $y \sim U[\frac{1}{6}, \frac{7}{6}]$. The scaling of x-axis is to reflect a measure of 1/3 students prefer c_2 as their top choice, while more students — a measure of 2/3 — prefer c_1 the most. In addition, the payoff of going to c_3 is normalized to $-1/6$. The white area represents students who submit $\pi_1 : c_1, c_2, c_3$ at equilibrium. Students inside the shaded region submit instead $\pi_2 : c_2, c_1, c_3$ at equilibrium. (2) In Subfigure (A), among students with $y \geqslant 1/2$ only those with a minimum score of 0.5 are accepted by c_1. The rest of students are admitted to c_2 if they have a score no less than 0.33, otherwise they are accepted by c_3. Students with $y < 1/2$ are accepted by c_2 if their scores are no less than 0.33. (3) In Subfigure (B), among students who submit π_1 ($y \geqslant 0.6$), only those with scores no less than 0.41 are accepted at c_1, the rest go to c_3; among students who submit π_2 ($y < 0.6$), those with minimum score 0.22 are admitted to c_2, the rest are admitted to c_3.

Figure 6: Expected utility under pre-exam submissions



Note: Preference intensity y lies in x-axis, higher value indicates stronger preference intensity for c_1. The solid line indicates the expected utility in the game induced by pre-exam submissions, and the dashed line presents the expected utility under post-score submissions. The example parameters are the same as in Figure 5.

(and manipulate their preference at equilibrium), while it hurts those students with an intermediate level of preference intensity as compared the post-score submissions. The introduction of heterogeneity suggests that some students who genuinely prefer c_2 top rank it anyhow, reducing the incentive to manipulate their top choices for those who prefer c_1. This has an additional effect in addition to a lower probability of getting into c_2, it also lowers the probability of getting into c_1 as less students will not top rank c_2 comparing to the homogenous case. Therefore, those with high enough preference intensity for c_1 can benefit under the pre-exam submissions. Similarly, those with sufficiently low preference intensity for c_1 find that manipulation in Boston yields a higher expected utility than under post-score submissions.